Steps to install TomCat Server

Step 1: Access Terminal and Update System

- **sudo -i** → Enter superuser mode to gain administrative control over the system.
- **sudo apt update && sudo apt upgrade -y** → Refreshes package indexes and upgrades installed software to the latest versions automatically.

```
vboxuser@prathwish03:~$ sudo -i
[sudo] password for vboxuser:
root@prathwish03:~# sudo apt update && sudo apt upgrade -y
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:5 https://dl.google.com/linux/chrome/deb stable InRelease
Ign:6 https://pkg.jenkins.io/debian binary/ InRelease
Get:7 https://pkg.jenkins.io/debian binary/ Release [2,044 B]
Get:8 https://pkg.jenkins.io/debian binary/ Release.gpg [833 B]
Ign:8 https://pkg.jenkins.io/debian binary/ Release.gpg
Reading package lists... Done
W: GPG error: https://pkg.jenkins.io/debian binary/ Release: The following signatures couldn't be verified because the public key is not available: NO_P
BKEY 7198F4B714ABFC68
E: The repository 'https://pkg.jenkins.io/debian binary/ Release' is not signed.
N: Updating from such a repository can't be done securely, and is therefore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configuration details.
root@prathwish03:~#
```

Step 2: Retrieve Apache Tomcat Package

- cd /tmp → Move to the temporary directory where downloads are usually stored.
- Use wget to fetch the Tomcat archive file from the official Apache repository:

wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz

```
root@prathwish03:/tmp# cd ..
root@prathwish03:/# cd /tmp
root@prathwish03:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-15 17:05:50--  https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.95.tar.gz'

apache-tomcat-9.0.95.tar.gz          100%[===================================================================>]  12.13M   343KB/s    in 46s

2026-04-15 17:06:40 (269 KB/s) - 'apache-tomcat-9.0.95.tar.gz' saved [12715996/12715996]
```

This downloads the compressed Tomcat package into your system.

Step 3: Prepare Installation Location

- cd /opt → Navigate to the directory used for optional third-party software.
- mkdir tomcat → Create a dedicated folder to hold Tomcat files.
- ls → Confirm that the new directory has been successfully created.

```
root@prathwish03:/tmp# cd ..
root@prathwish03:/# cd /opt
root@prathwish03:/opt# ls
google  VBoxGuestAdditions-7.2.6
root@prathwish03:/opt# mkdir tomcat
root@prathwish03:/opt# ls
google  tomcat  VBoxGuestAdditions-7.2.6
root@prathwish03:/opt#
```

Step 4: Unpack the Tomcat Archive

- Extract the downloaded file into the created directory:

tar -xvzf /tmp/apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1

```
root@prathwish03:/# cd /tmp
root@prathwish03:/tmp# sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=
apache-tomcat-9.0.95/conf/
apache-tomcat-9.0.95/conf/catalina.policy
apache-tomcat-9.0.95/conf/catalina.properties
apache-tomcat-9.0.95/conf/context.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xml
apache-tomcat-9.0.95/conf/jaspic-providers.xsd
apache-tomcat-9.0.95/conf/logging.properties
apache-tomcat-9.0.95/conf/server.xml
apache-tomcat-9.0.95/conf/tomcat-users.xml
apache-tomcat-9.0.95/conf/tomcat-users.xsd
apache-tomcat-9.0.95/conf/web.xml
apache-tomcat-9.0.95/bin/
apache-tomcat-9.0.95/lib/
apache-tomcat-9.0.95/logs/
apache-tomcat-9.0.95/temp/
apache-tomcat-9.0.95/webapps/
apache-tomcat-9.0.95/webapps/ROOT/
apache-tomcat-9.0.95/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.95/webapps/docs/
apache-tomcat-9.0.95/webapps/docs/META-INF/
apache-tomcat-9.0.95/webapps/docs/WEB-INF/
apache-tomcat-9.0.95/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.95/webapps/docs/annotationapi/
apache-tomcat-9.0.95/webapps/docs/api/
apache-tomcat-9.0.95/webapps/docs/appdev/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/docs/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/src/
apache-tomcat-9.0.95/webapps/docs/appdev/sample/src/mypackage/
```

This removes the root folder and directly places all files into /opt/tomcat.

Step 5: Modify Server Configuration

- cd /opt/tomcat/conf → Enter the configuration folder.

```
root@prathwish03:/tmp# cd ..
root@prathwish03:/# cd /opt
root@prathwish03:/opt# cd tomcat/
root@prathwish03:/opt/tomcat# ls
bin  BUILDING.txt  conf  CONTRIBUTING.md  lib  LICENSE  logs  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  temp  webapps  work
root@prathwish03:/opt/tomcat# cd conf
root@prathwish03:/opt/tomcat/conf# ls
catalina.policy      context.xml          jaspic-providers.xsd  server.xml        tomcat-users.xsd
catalina.properties  jaspic-providers.xml  logging.properties    tomcat-users.xml  web.xml
root@prathwish03:/opt/tomcat/conf# gedit server.xml
```

- Edit the main configuration file:

nano server.xml

- Change connector port:

<Connector port="9090" ... />

```
<!-- A "Service" is a collection of one or more "Connectors" that share
     a single "Container" Note:  A "Service" is not itself a "Container",
     so you may not define subcomponents such as "Valves" at this level.
     Documentation at /docs/config/service.html
 -->
<Service name="Catalina">

  <!--The connectors can use a shared executor, you can define one or more named thread
ols-->
  <!--
  <Executor name="tomcatThreadPool" namePrefix="catalina-exec-"
      maxThreads="150" minSpareThreads="4"/>
  -->


  <!-- A "Connector" represents an endpoint by which requests are received
       and responses are returned. Documentation at :
       Java HTTP Connector: /docs/config/http.html
       Java AJP  Connector: /docs/config/ajp.html
       APR (HTTP/AJP) Connector: /docs/apr.html
       Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
  -->
  <Connector port="9090" protocol="HTTP/1.1"
             connectionTimeout="20000"
             redirectPort="8443"
             maxParameterCount="1000"
             />
  <!-- A "Connector" using the shared thread pool-->
  <!--
  <Connector executor="tomcatThreadPool"
             port="8080" protocol="HTTP/1.1"
             connectionTimeout="20000"
             redirectPort="8443"
             maxParameterCount="1000"
             />
```

This avoids conflicts if port 8080 is already in use.

Step 6: Set Up User Authentication

- Open the user configuration file:

nano tomcat-users.xml

- Add a user with appropriate roles:

<user username="admin" password="admin123" roles="manager-gui,admin-gui"/>

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
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  contributor license agreements.  See the NOTICE file distributed with
  this work for additional information regarding copyright ownership.
  The ASF licenses this file to You under the Apache License, Version 2.0
  (the "License"); you may not use this file except in compliance with
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  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
              xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
              xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
              version="1.0">
              <user username="admin" password="Admin" roles="manager-gui">
<!--
  By default, no user is included in the "manager-gui" role required
  to operate the "/manager/html" web application.  If you wish to use this app,
  you must define such a user - the username and password are arbitrary.

  Built-in Tomcat manager roles:
    - manager-gui    - allows access to the HTML GUI and the status pages
    - manager-script - allows access to the HTTP API and the status pages
    - manager-jmx    - allows access to the JMX proxy and the status pages
    - manager-status - allows access to the status pages only
```

This enables access to the management dashboard.

Step 7: Launch Tomcat Server

- Navigate to binary scripts:

cd /opt/tomcat/bin

- Start the server:

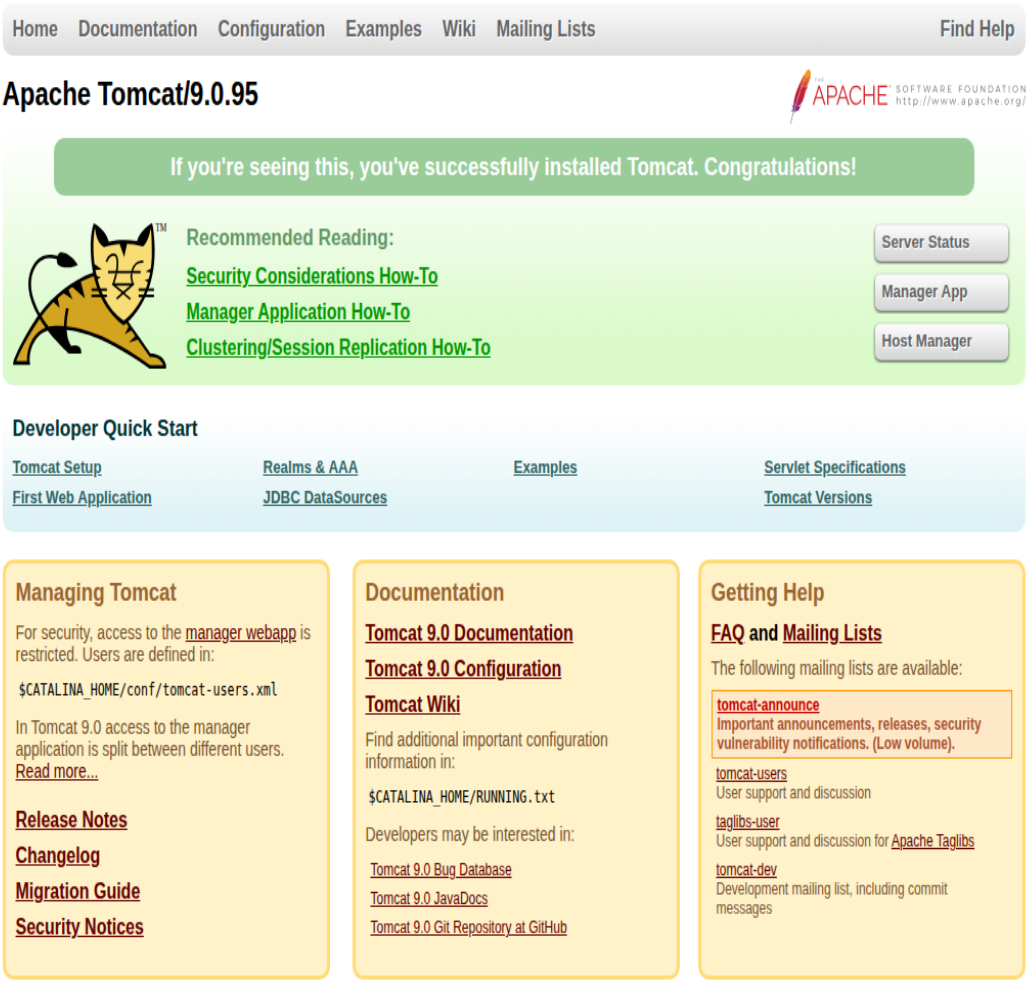
sh startup.sh

```
root@prathwish03:/opt/tomcat/conf# cd ..
root@prathwish03:/opt/tomcat# cd bin
root@prathwish03:/opt/tomcat/bin# ls
bootstrap.jar       ciphers.bat                 configtest.bat  digest.sh        setclasspath.sh  startup.sh           tool-wrapper.sh
catalina.bat        ciphers.sh                  configtest.sh   makebase.bat     shutdown.bat     tomcat-juli.jar      version.bat
catalina.sh         commons-daemon.jar          daemon.sh       makebase.sh      shutdown.sh      tomcat-native.tar.gz version.sh
catalina-tasks.xml  commons-daemon-native.tar.gz digest.bat      setclasspath.bat startup.bat      tool-wrapper.bat
root@prathwish03:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@prathwish03:/opt/tomcat/bin#
```

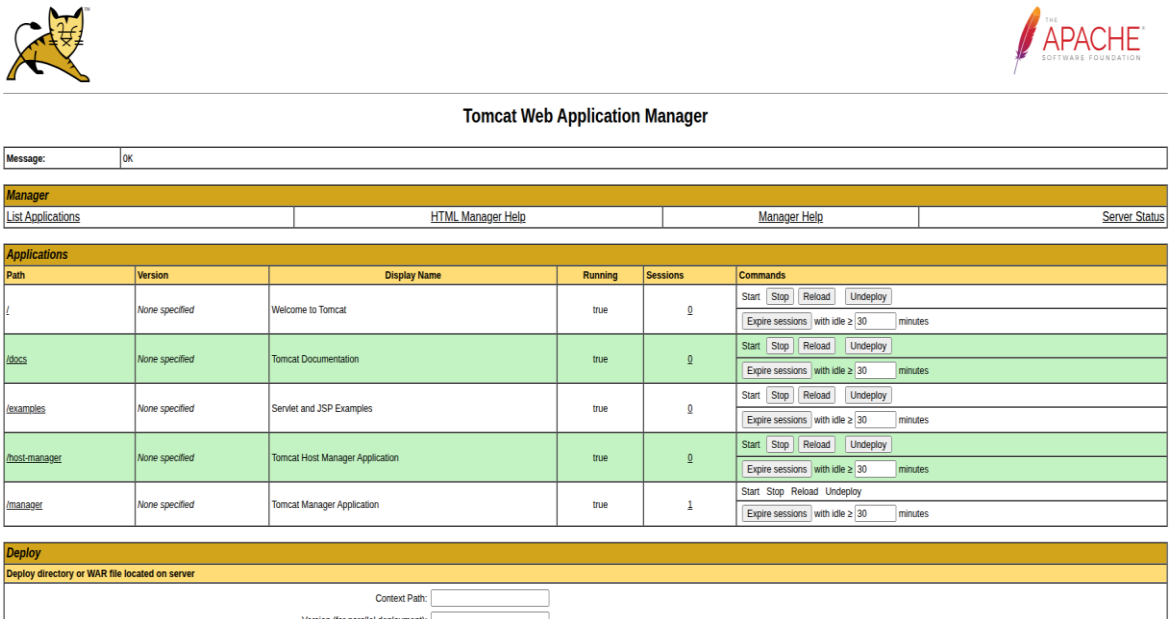
Tomcat service will start in the background.

Step 8: Confirm Successful Deployment

- Open browser and visit:
    - http://localhost:9090 → Displays Tomcat homepage



- http://localhost:9090/manager/html → Opens admin panel



Tomcat Web Application Manager

Message:	OK

Manager

List Applications	HTML Manager Help	Manager Help	Server Status

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	*None specified*	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	*None specified*	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	*None specified*	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	*None specified*	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	*None specified*	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Enter credentials to access deployment tools.

MymavenWebApp01 pipeline

Step 1: Create new file named "**Jenkinsfile**" inside MymavenWebApp01 folder and push into git

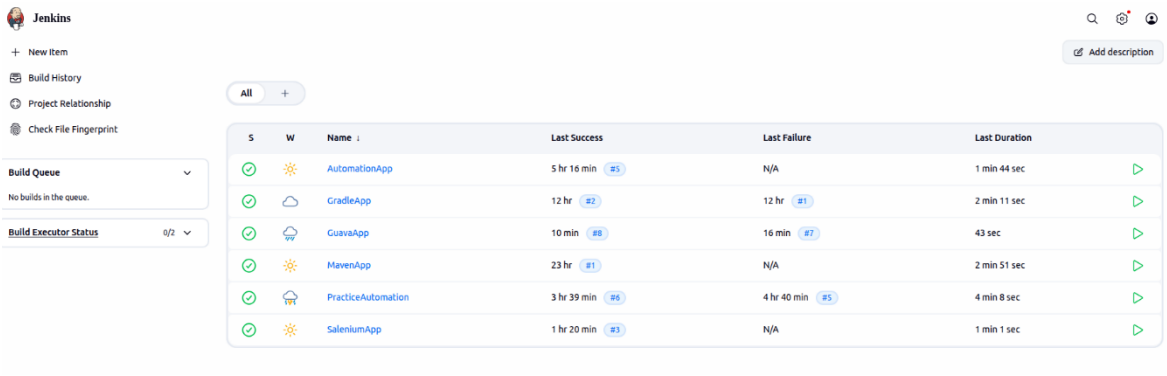
Jenkinsfile:

```
pipeline {
   agent any

   tools {
      maven 'Maven'   // Must match Jenkins tool name
   }

   stages {

      stage('Checkout') {
         steps {
            git branch: 'master', url: 'https://github.com/PrathwishShetty03/MymavenWebApp01.git'
         }
      }

      stage('Build') {
         steps {
            sh 'mvn clean package'
         }
      }

      stage('Test') {
         steps {
            sh 'mvn test'
         }
      }

      stage('Deploy WAR to Tomcat') {
         steps {
            sh '''
            echo "Stopping Tomcat..."
            /opt/tomcat/bin/shutdown.sh || true
            sleep 5

            echo "Removing old WAR..."
            rm -rf /opt/tomcat/webapps/MymavenWebApp01*

            echo "Copying new WAR..."
            cp target/MymavenWebApp01.war /opt/tomcat/webapps/

            echo "Starting Tomcat..."
            /opt/tomcat/bin/startup.sh
```

```
            '''
        }
    }
}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}
    stage('Build') {
        steps {
            sh 'mvn clean package -DskipTests'
        }
    }

    stage('Run Application') {
        steps {
            sh '''
            echo "Running using Maven..."
            mvn exec:java -Dexec.mainClass="com.example.App"
            '''
        }
    }
}

post {
    success {
        echo ' Build and execution SUCCESSFUL!'
    }
    failure {
        echo ' Build FAILED!'
    }
}
}
```

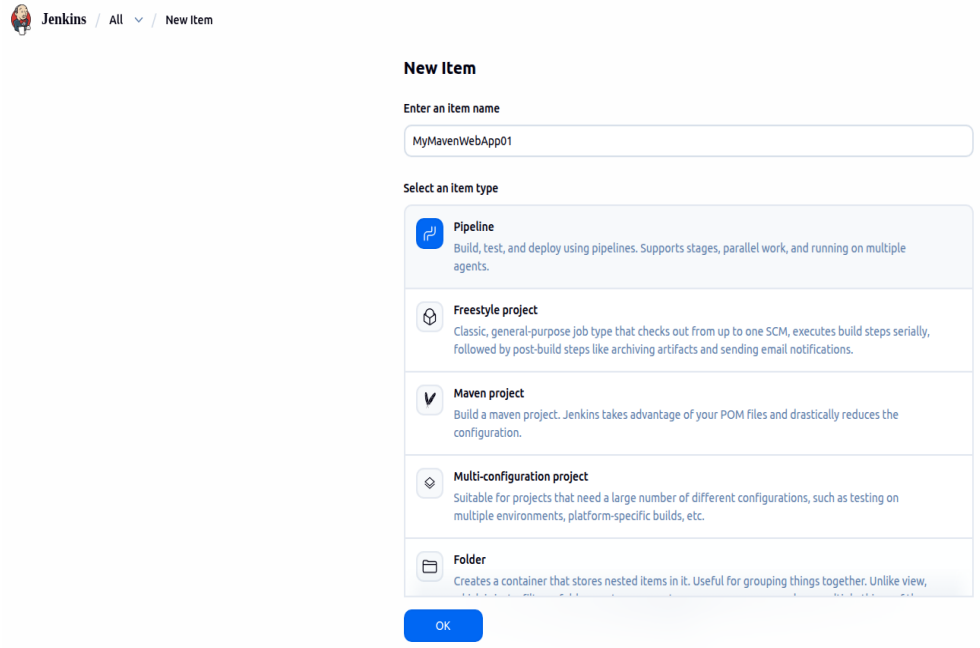
Step 2:Open Jenkins Dashboard

- Open Jenkins in browser
- Click on New Item



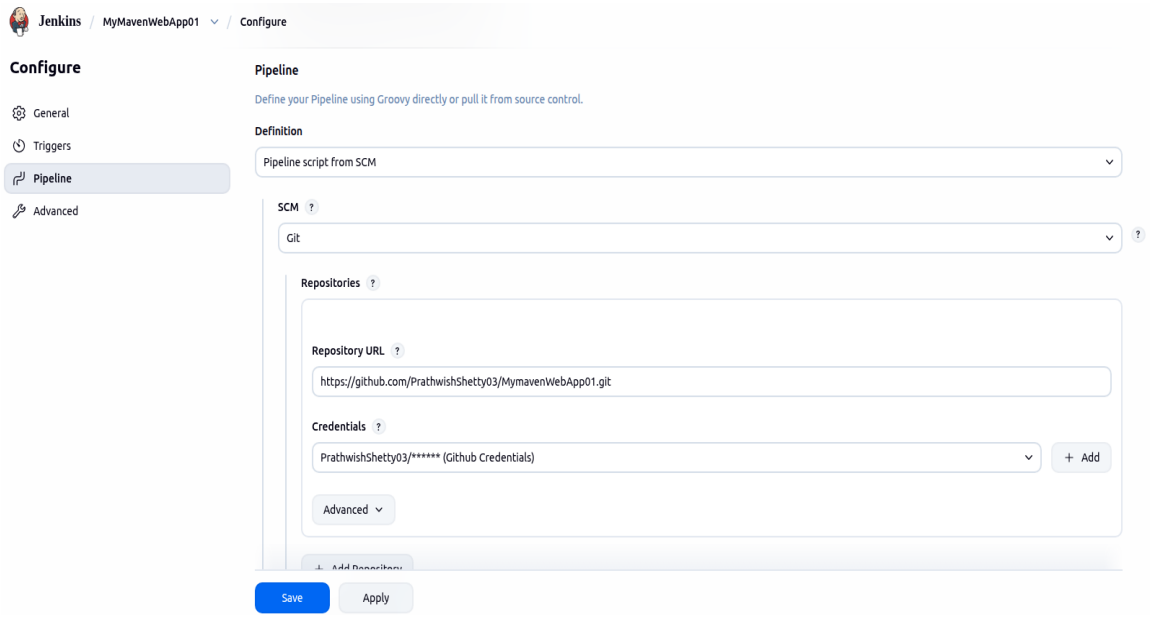
Step 3: Create Pipeline Job

- Enter name: MymavenWebApp01
- Select **Pipeline**
- Click **OK**



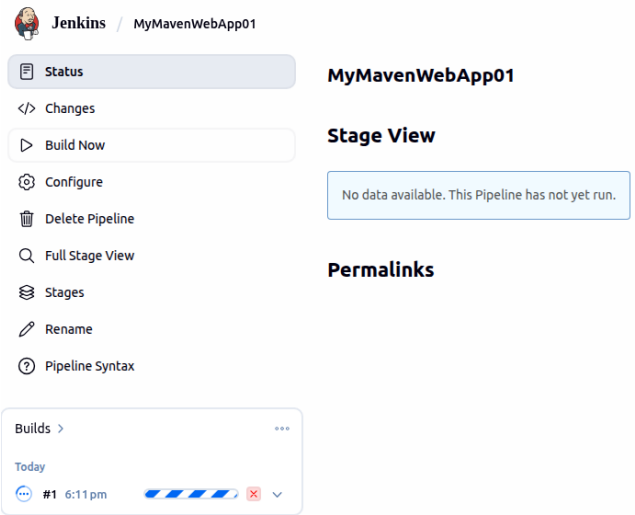
Step 4: Configure Pipeline with GitHub

- Select **Pipeline script from SCM**
- SCM → Git
- Enter repository URL
- Add credentials
- Click **Save**



Step 5: Run the Pipeline

- Click **Build Now**



Step 6: Look for the status of project

- Check Stage View
- View Build History